GENERAL INFORMATION

DESCRIPTION AND OPERATION

IMAGE PROCESSING CONTROL MODULE (IPCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Auto High Beam Control Module (AHBCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.
For additional information, refer to: Headlamps (417-01 Exterior Lighting,

Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1286-16	Interior Mirror - Circuit voltage below threshold	▪ Image processing control module power or ground circuit open circuit, high resistance ▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - ECU Power Supply Voltage (0xD112). Refer to the electrical circuit diagrams and check the image processing control module power and ground circuits for open circuit, high resistance ▪ Refer to the relevant section of the workshop manual and test the battery and charging system
B1286-17	Interior Mirror - Circuit voltage above threshold	▪ Battery/charging system fault	▪ Refer to the relevant section of the workshop manual and test the battery and charging system
B1286-44	Interior Mirror - Data memory failure	▪ Image processing control module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module
B1286-47	Interior Mirror - Internal electronic failure	▪ Image processing control module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module
B1286-49	Interior Mirror - Internal electronic failure	▪ Image processing control module active light sensor failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module
B1286-60	Interior Mirror - Reserved By Document	▪ Image processing control module temperature below operational limits	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module

B1286-78	Interior Mirror - Alignment or adjustment incorrect	▪ Image processing control module operating mode has been changed	▪ Using the manufacturer approved diagnostic system, clear the DTCs, cycle ignition and retest. If the fault persists, install a new image processing control module
B1286-96	Interior Mirror - Component internal failure	▪ Image processing control module internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module
B1286-97	Interior Mirror - Component or system operation obstructed or blocked	▪ Image processing control module camera obstructed	▪ Remove obstructions from in front of the image processing control module
B1286-98	Interior Mirror - Component or system over temperature	▪ Image processing control module temperature above operational limits	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new image processing control module
B12AC-11	Electrochromatic Door Mirror Output - Circuit short to ground	▪ No door mirrors electrochromatic function ▪ Door mirrors electrochromatic circuit(s) short circuit to ground	▪ Refer to electrical circuit diagrams and check the door mirror electrochromatic circuit(s)for short circuit to ground. Repair circuit as required, clear DTC and retest
C1001-06	Vision System Camera - Algorithm based failures	▪ Image processing control module camera misaligned	▪ Align the image processing control module camera. Clear the DTC and retest
C1001-78	Vision System Camera - Alignment or adjustment incorrect	▪ Image processing control module camera misaligned	▪ Align the image processing control module camera. Clear the DTC and retest

U0001-88	High Speed CAN Communication Bus - Bus off	▪ High speed CAN bus (chassis) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the high speed CAN bus (chassis) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0002-81	High Speed CAN Communication Bus Performance - Invalid serial data received	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0002-82	High Speed CAN Communication Bus Performance - Alive/sequence counter incorrect / not updated	▪ Invalid data received from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0002-87	High Speed CAN Communication Bus Performance - Missing message	▪ Missing message from another control module via the high speed CAN bus (chassis)	▪ Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ Car configuration file mismatch with vehicle specification	▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U201A-57	Control Module Main Calibration Data - Invalid / incomplete	▪ Image processing control module is not	▪ Using the manufacturer approved diagnostic system, reconfigure the image processing control module with the latest level

	software component	configured correctly	software
U2300-54	Central Configuration - Missing calibration	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid/incomplete configuration	▪ Car configuration file mismatch with vehicle specification	**NOTE:** After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs ▪ Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and

			retest